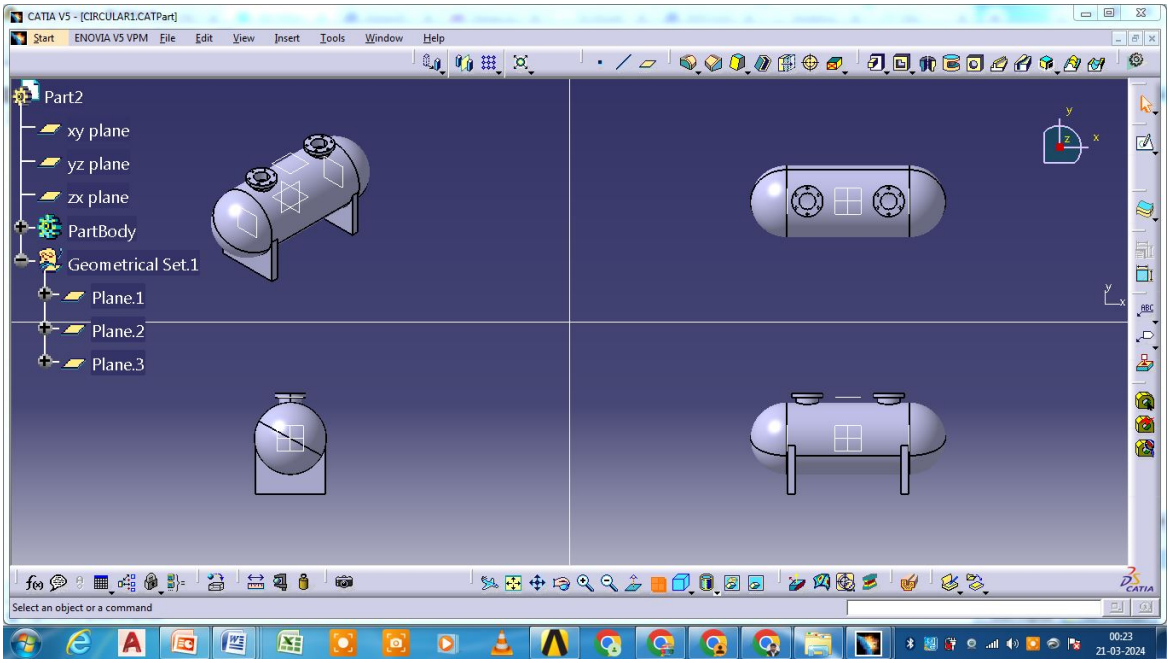
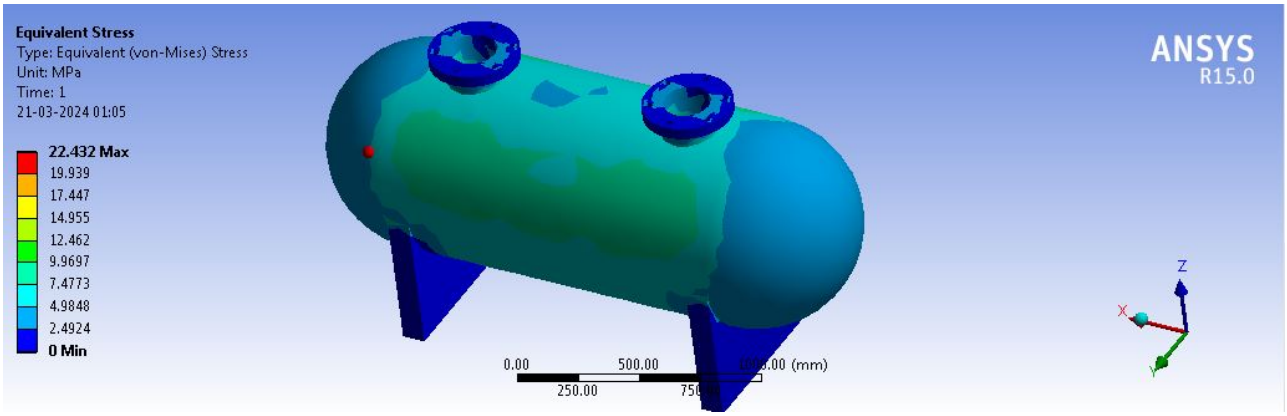
Pressure vessel: native CAD and saved FEM

12 October 2026. Three new deformable-FEM families collected: nozzle-and-saddle pressure vessel, perforated brake disc and double-wishbone assembly. A fourth robot-arm archive is retained separately for rigid joint motion. No generated CAD or new solver runs.



Original publisher CAD views: vessel, nozzle openings, heads and saddle supports. Native CATIA parts and three IGES exports retained. CIRCULAR and CIRCULAR1 have matching imported geometry extents and 66 faces each, but different original hashes. FLAT HEAD imports with 72 faces. None imports as a watertight solid; source native parts remain available.

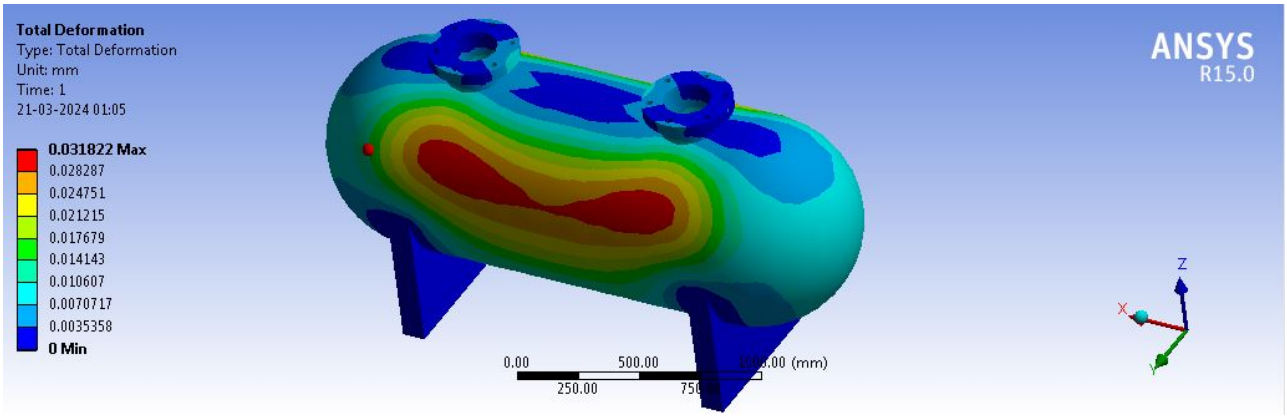


Original FEM figure: Nimonic 75 equivalent stress, maximum 22.432 MPa. Raw file.rst and complete static input deck are available. This is one saved material/head case, rather than a solved case for every screenshot folder.

CAD scale: circular-head IGES extent approximately 2400 x 880 x 1239.90 mm. The views have no verified complete dimensioned engineering drawing. Supplied-CAD FEM and drawing reconstruction remain separate benchmark tracks.

Source: original publisher repository. Pinned revision ac46e4eed4; download hashes and source notices retained.

Pressure vessel: numerical reference coverage



Original deformation figure: maximum 0.031822 mm. Read-only extraction agrees at the printed precision; source arrays are retained unchanged in saved-static-fields.npz.

Reference audit	Saved result / source setup
Mesh and state	8,709 nodes; 4,965 elements; one static state labeled 1
Maximum translation	0.031822459820 mm
Finite recovered maximum VM stress	22.431762698 MPa
Finite stress rows	1,462; missing rows 7,247
Source material	Nimonic 75; E = 207,000 MPa; Poisson ratio 0.34
Pressure and units	0.824 MPa; N-mm-tonne-s, original /units,MPA

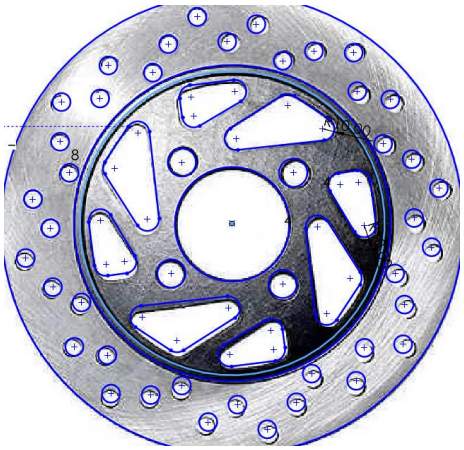
Stress interpretation: ansys-mapdl-reader 0.56.0 averages saved element-nodal stress at shared corner nodes. The reported VM maximum uses finite six-component recovered rows. Original NaNs, node IDs and coordinates are preserved. Do not compare against a differently recovered or integration-point stress field.

Admission limits: CAD/result head revision and contact/support details need full mapping and mesh-convergence review. README volume claim of 24 liters conflicts with its listed 880 mm outside diameter, 20 mm wall and 1520 mm shell length. The saved case can still support a source-setup benchmark after this mismatch is documented.

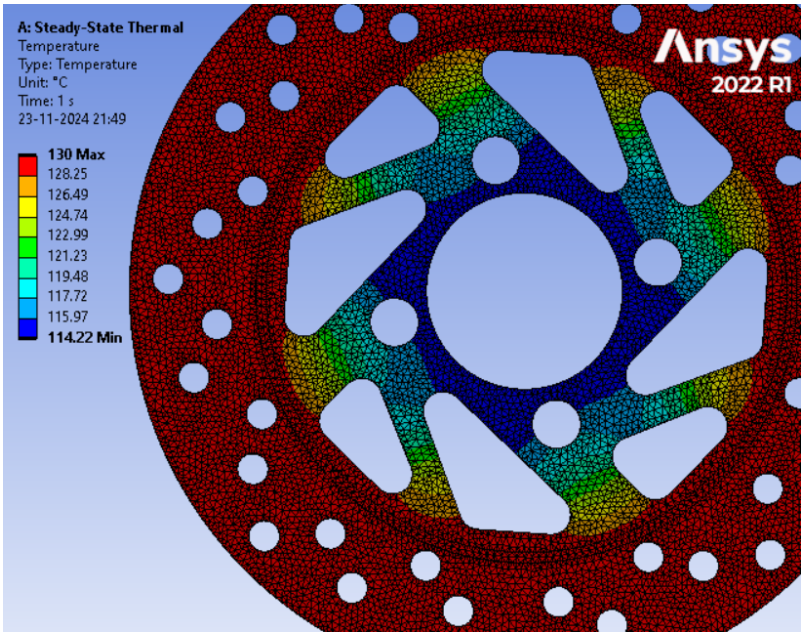
The saved RST is a static pressure solution. Published thermal tables/plots do not establish a corresponding saved thermal field. Material and head variants in folders are not additional validated raw cases.

Source: original publisher repository. Pinned revision ac46e4eed4; download hashes and source notices retained.

Perforated brake disc: CAD and thermal FEM



Original publisher modeling sketch: brake-disc geometry traced from a physical disc. This is a CAD construction view, not a complete dimensioned manufacturing drawing. Native SLDPRT, valid STEP and DesignModeler AGDB retained.



Original FEM figure: steady-state temperature, source report physical page 4. Legend 114.22 to 130 degrees C. The saved thermal RTH is finite at all 188,762 mesh nodes; mesh has 119,885 elements.

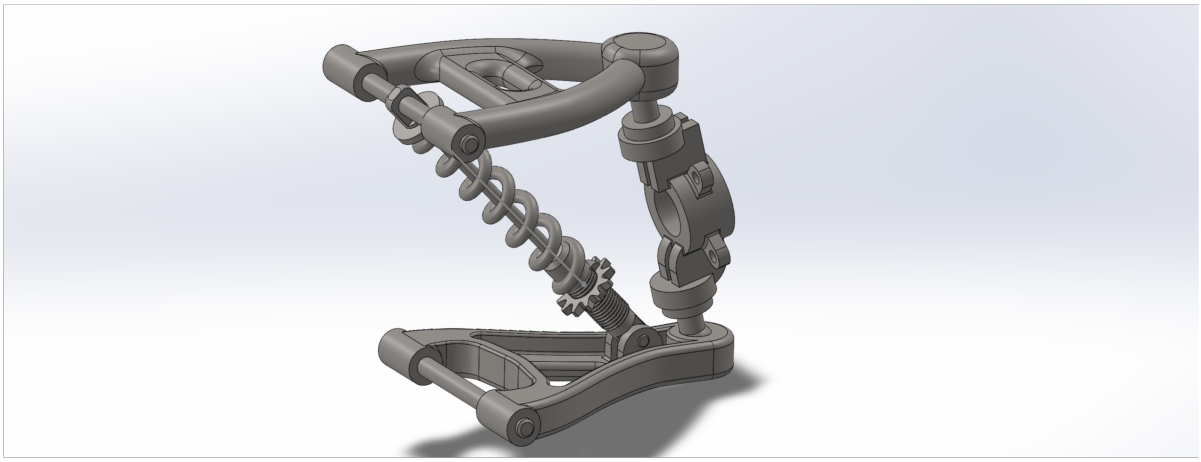
Saved thermal reference	Value
Minimum temperature	114.216616940 degrees C
Maximum temperature	130.000004877 degrees C
Boundary conditions	Prescribed 130 degrees C; ambient 22 degrees C; temperature-dependent convection table

saved-thermal-fields.npz contains node IDs, coordinates and Celsius temperature read from the original RTH. The thermal nodal solution is a scalar temperature, despite the generic reader label. Peak temperature is prescribed; field/interior error is a more useful potential metric.

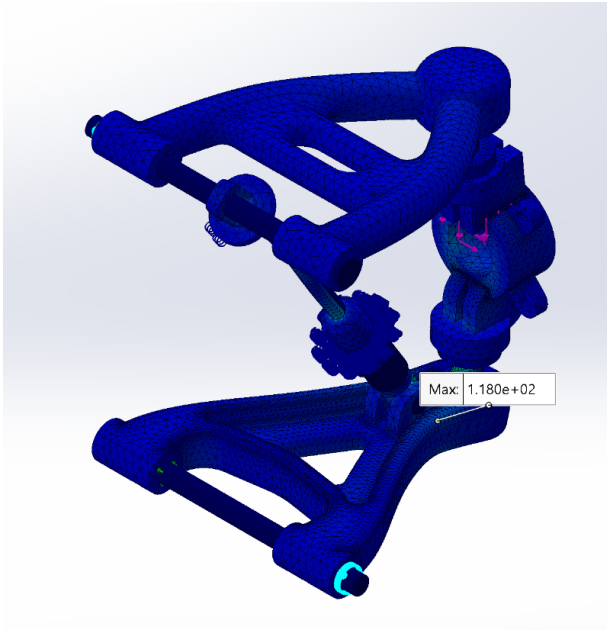
No raw structural RST found. Structural report and topology outputs are auxiliary references. Fluent CFD assets use finite volume and are excluded from FEM counts. Exact CAD/node mapping and convergence still require review.

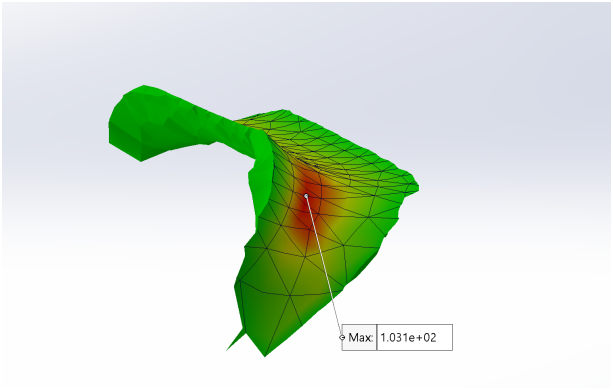
Source: original publisher repository. Pinned revision 508a55937d; download hashes and source notices retained.

Double wishbone: complex assembly and FEA



Original CAD illustration: wishbones, upright, spring, pins and geared actuator. Native SolidWorks assembly plus nine part files and neutral STEP are retained. Independent STEP import is valid: 10 solids, 549 faces, approximately 257.81 x 324.58 x 220.00 mm.





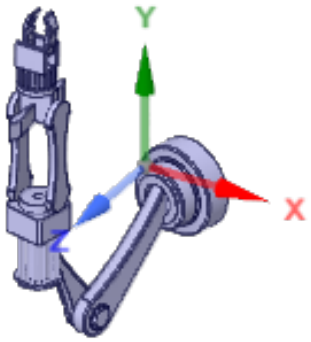
Original source FEM figures: baseline callout 118 MPa (left); redesigned-region callout 103.1 MPa (right). The report prose rounds the latter to 103 MPa. These are published scalar references; raw mesh and saved simulation fields were explicitly omitted.

Source setup / reference	Value
Applied force components	3675 N vertical; 375 N longitudinal
Baseline / redesigned maximum stress	118 MPa / 103.1 MPa (image precision)
Connector and fixtures	Pin/hinge constraints; spring replaced by a connector

Before/after CAD revision matching is unresolved. The report contains yield-strength placeholders and an unusual spring-stiffness normalization, so an exact FoS or a fully reproducible setup is not claimed. Simulink/Simscape control results are kept as context and are not FEM cases.

Source: original publisher repository. Pinned revision 3f2dd9ecde; download hashes and source notices retained.

Robot arm: native CAD, rigid-motion archive



Original CAD thumbnail: articulated robot arm in source SpaceClaim. The downloaded project ZIP contains a native .scdocx with eight nonempty serialized Parasolid geometry members, source Mechanical database, input deck and saved RST. Independent native BREP validity has not been imported.

Saved source inventory	Interpretation
59 nodes, 43 elements	Reference nodes, masses, targets and joints
480 saved states	One trajectory, source labels 1 to 480
Element formulations	MASS21, TARGE170, MPC184
Source input	All nine bodies sent as rigid; transient, large-motion MKS setup
Stored motion array	480 x 59 x 6 generalized displacement, NaNs preserved
Continuum stress reference	Absent: no ENS stresses in saved result

Held from deformable-FEM benchmark: the archive supplies actual CAD and saved joint-motion output. Its coarse reference-node representation and lack of solid/shell stress do not support a continuum stress benchmark. The source README stress description is broader than the stored fields.

saved-rigid-motion-only.npz is a read-only extraction for possible future joint/multibody tasks. Rotations, translational coordinates, reference nodes and missing entries need explicit mapping. Never count 480 frames as 480 independent engineering tasks or treat rigid-body motion as elastic deformation.

Input explicitly says antype,4 transient, while the binary header analysis code is 0. Preserve both metadata signals; resolve semantics from source input and output logs before any trajectory scoring.

Source: original publisher repository. Pinned revision 932111edef; download hashes and source notices retained.

New collection: counts and benchmark status

Family	Original CAD + FEM evidence	Candidate status
Pressure vessel	CATIA + IGES; RST, input deck, stress/deformation figures	One saved static FEM candidate
Brake disc	SLDPRT + valid STEP + AGDB; scalar-temperature RTH	One saved thermal FEM candidate
Double wishbone	SolidWorks + valid STEP; original FEM scalar figures	One plot-reference candidate
Robot arm	SpaceClaim + saved rigid joint-motion RST	Auxiliary; excluded from continuum stress

Counts: three deformable-FEM geometry families: two native-CAD/saved-field candidates and one native-CAD/plot-reference candidate. One additional rigid-motion family is held separately. Two independently imported valid STEP models; pressure IGES has valid faces without closed solids. Zero newly verified complete dimensioned engineering drawings and zero certified admissions.

Before scoring: verify exact CAD/result revision matching, dimensions, material scope, loads/constraints, numerical convergence, node correspondence and field-recovery conventions. Missing stress entries stay missing. Original plotted maxima are engineering reference values, not measured frontier-model scores.

Two task tracks: supplied-CAD-to-FEM can use admitted candidates. Drawing-to-CAD-to-FEM requires sufficient original dimensioned drawings; screenshots, thumbnails and modeling traces do not automatically meet that requirement.

Collection integrity: pinned source revisions, original source paths/URLs, byte lengths and SHA-256 hashes accompany downloads. Source code and input decks were read as text. No new CAD, solver run or model call occurred. No COMSOL files were used.

Local research: explicit source dataset licenses were not found. Original notices retained; no public redistribution. Case-sensitive pressure-vessel folders were stored with a documented alias to preserve distinct originals on this filesystem.

Machine-readable catalogs provide detailed source paths and limitations. The master dataset index registers this extension without counting material screenshots, frames, submodel files or node rows as extra independent cases.